

Reproducibility Statement for Dual-Branch Synergistic Learning for Coordinated Action Quality Assessment in Pair Skating

1 Model Overview

The proposed DBSModel predicts pair skating competition scores by integrating visual, motion, audio, and skater-specific features. Each modality is projected into a shared 256-dimensional feature space. Leader-Follower Coordination (LFC) modules model interactions between skaters, followed by multi-modal fusion and hierarchical temporal convolution blocks. A lightweight prediction head outputs the final score.

1.1 Input Modalities

Table 1: Input modalities and feature dimensions.

Modality	Feature Dimension
RGB Visual Features	1024
Optical Flow Features	1024
Audio Features	768
Skater 1 Features	1024
Skater 2 Features	1024

Output: Scalar score in the range $[0, 1]$.

2 Dataset

The proposed PairSkate dataset contains pair skating competition videos with score-level and element-level annotations.

Table 2: Dataset statistics.

Subset	Videos
Training Set	409
Validation/Test Set	104

Features are extracted using VST (RGB), I3D (Optical Flow), AST (Audio), and custom skater-centric visual encoders.

3 Training Configuration

Table 3: Training configuration.

Parameter	Value
Optimizer	AdamW
Learning Rate	5×10^{-4}
Weight Decay	1×10^{-4}
Scheduler	Cosine Annealing
Batch Size	32
Epochs	500
Feature Dimension	256
Attention Heads	4
Coordination Loss Weight	0.2
Rhythm Loss Weight	0.2
Feature Dropout	0.4
Head Dropout	0.3

4 Evaluation Metrics

4.1 Spearman Correlation

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \quad (1)$$

where d_i denotes the rank difference between prediction and ground truth.

4.2 Relative L2 Distance (R-L2)

$$R-L2 = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2}}{\frac{1}{N} \sum_{i=1}^N |y_i|} \quad (2)$$

where $\hat{y}_i$ and y_i denote the predicted and ground-truth scores, respectively.

5 Code and Data Availability

- The codebase, including model definitions, training scripts, evaluation scripts, and configuration files, will be publicly released.
- Dataset preprocessing scripts are provided.
- The PairSkate dataset will be publicly available for research purposes.

6 Reproduction Instructions

6.1 Environment Setup

```
# Create conda environment
conda create -n dbs python=3.8 -y
```

```
conda activate dbs

# Install PyTorch (CUDA 11.8)
pip install torch torchvision torchaudio \
--index-url https://download.pytorch.org/whl/cu118

# Install dependencies
pip install -r requirements.txt
```

6.2 Train

```
# Train TES model
python train.py --gpu 0 --score_type tes_score

# Train PCS model
python train.py --gpu 0 --score_type pcs_score
```

6.3 Test

```
# Test TES model
python test.py --gpu 0 \
--ckpt_path checkpoints/dbs_v22/dbs_tes_score_best.pth \
--score_type tes_score

# Test PCS model
python test.py --gpu 0 \
--ckpt_path checkpoints/dbs_v22/dbs_pcs_score_best.pth \
--score_type pcs_score
```

7 Hardware Requirements

- GPU: NVIDIA RTX 3090 or higher
- GPU Memory: at least 8 GB
- System Memory: at least 16 GB
- Storage: at least 50 GB